Linards Balodis

Sigulda, Latvia | **Portfolio:** linardsb.xyz | linardsbalodis2009@gmail.com | github.com/Linards888 | linkedin.com/in/linards-balodis | instagram.com/linards888

Engineering student — robotics, electronics and mechanical design. I build projects from the first sketch to a working product.

Education

Sigulda State Gymnasium, Sigulda — *2022 – present*

General secondary education, engineering track — Average grade: 7.5 / 10

Strongest subjects: physics, chemistry, mathematics | Activities: Young Physicists' School (JFS), mathematics, physics and chemistry olympiads

Projects & Achievements

LATVIAN ROBOTICS CHAMPIONSHIP | *Multiple podium finishes* — *2022 – present*

- Combat robots, 1 lb class — multiple podium finishes at Latvian championship stages.
- Folkrace autonomous racing — multiple podium finishes at Latvian championship stages. Built the robot with sensors and control logic.
- Top 16 in Europe — RoboChallenge 2025 and 2026 international competitions.

PCB DESIGN, MODELLING AND SOLDERING | *Personal projects* — *2026 – present*

- Designing a flight computer for a self-built rocket with an E-class rocket motor.
- Designing a compact PCB for combat robots with an integrated receiver, brushless motor driver and more.

RESEARCH PROJECT (ZPD) | *Scientific research paper* — *2026 – present*

- Researching image translation methods for spiking neural networks (SNNs).

VENTSPILS INTERNATIONAL COMBAT ROBOT HACKATHON | *Combat Robotics* — *2024*

- Competed in an international combat robot competition against experienced teams.
- Built the robot myself: KiCad PCB, Fusion 360 chassis and C++ control code.

DIY SIM RACING + FPV FOLKRACE ROBOT | *Team project* — *2024 – 2025*

- Built a sim racing rig with a DIY steering wheel and pedals, linked to an FPV Folkrace robot for remote real-time driving.
- Integrated a camera and video transmitter and optimised latency for real-time control.

LATA OPEN DATA HACKATHON & IDEA GENERATOR | *3rd place, Idea Generator* — *2023*

- Developed and publicly defended the idea "Improving public health in Latvia" using government open data.

PORTFOLIO WEBSITE | *Developer — linardsb.xyz* — *2024 – present*

- Designed, built and deployed my own Next.js website with a project portfolio and resume section.

Work Experience

GOODS RECEIVING | *Gāles nams, receiving department* — *2026*

- Receiving and checking deliveries — accuracy, responsibility and reliability in daily work.

WAITER | *Hospitality staff* — *2025*

- Communicating with customers under pressure, teamwork and problem-solving — skills that transfer to running engineering projects.

KINDERGARTEN ASSISTANT | *Teacher's assistant* — *2023*

- Responsibility, patience and the ability to explain complex things simply — valuable skills in technical projects too.

Skills

Programming	C++ · PlatformIO · Arduino · Python · Next.js / React · HTML · CSS · Git / GitHub
CAD & 3D	Fusion 360 · Onshape · 3D printing · Antweight & Folkrace mechanical design
Electronics	Soldering (SMD & through-hole) · PCB design (KiCad, Eagle) · Microcontrollers (ESP32, AVR, ATmega, Teensy) · Sensor integration · Robot control systems
Manufacturing	CNC milling on a 3-axis router with UGS (Universal G-code Sender) · G-code / CAM programming · Lathe turning · Aluminium and plastic machining
Languages	Latvian (native) · English (B2) · Russian (A1, limited)